

MYSQL – CLAUSES AND JOINS

Inserting the datas

The screenshot shows the MySQL Workbench interface with the following components:

- Navigator:** Shows the 'employee' database selected, with tables like departments, employees, and location visible.
- SQL Editor:** Contains the following SQL script:

```
1 USE employee;
2
3 INSERT INTO departments (department_id, department_name) VALUES
4 (1, 'Software Development'),
5 (2, 'Marketing'),
6 (3, 'Data Science'),
7 (4, 'Human Resources'),
8 (5, 'Product Management'),
9 (6, 'Content Creation'),
10 (7, 'Finance'),
11 (8, 'Design'),
12 (9, 'Research and Development'),
13 (10, 'Customer Support'),
14 (11, 'Business Development'),
15 (12, 'IT'),
16 (13, 'Operations');
17
18 INSERT INTO location (location) VALUES
19 ('Chennai'),
20 ('Bangalore'),
21 ('Hyderabad'),
22 ('Pune');
```
- Output:** Shows the execution results of the queries:

#	Time	Action	Message	Duration / Fetch
1	09:33:03	USE employee	0 row(s) affected	0.000 sec
2	09:33:37	INSERT INTO departments (department_id, department_name) VALUES (1, 'Software Development'), (2, ...	13 row(s) affected Records: 13 Duplicates: 0 Warnings: 0	0.016 sec
3	09:34:08	INSERT INTO location (location) VALUES ('Chennai'), ('Bangalore'), ('Hyderabad'), ('Pune')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.016 sec
- Context Help:** Displays a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

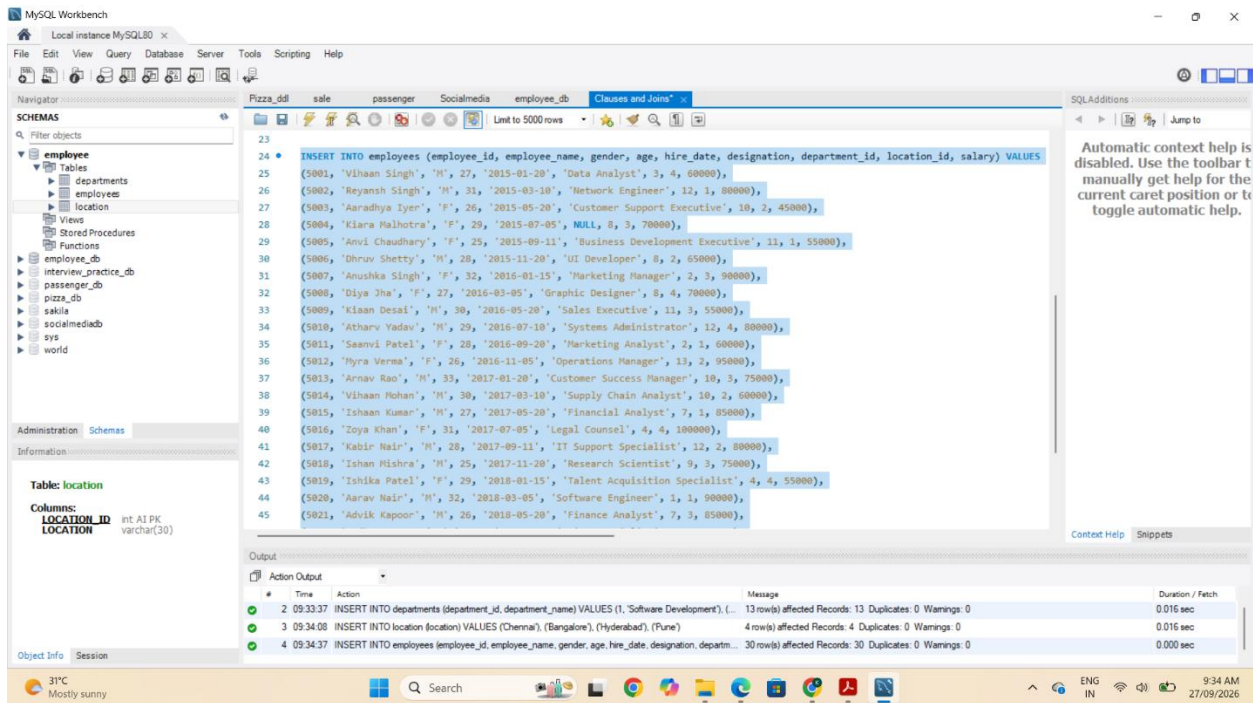
This screenshot shows the same MySQL Workbench interface, but with the SQL Editor displaying only the first two queries from the previous screenshot:

```
1 USE employee;
2
3 INSERT INTO departments (department_id, department_name) VALUES
4 (1, 'Software Development'),
5 (2, 'Marketing'),
6 (3, 'Data Science'),
7 (4, 'Human Resources'),
8 (5, 'Product Management'),
9 (6, 'Content Creation'),
10 (7, 'Finance'),
11 (8, 'Design'),
12 (9, 'Research and Development'),
13 (10, 'Customer Support'),
14 (11, 'Business Development'),
15 (12, 'IT'),
16 (13, 'Operations');
```

The Output panel shows the results for these two queries:

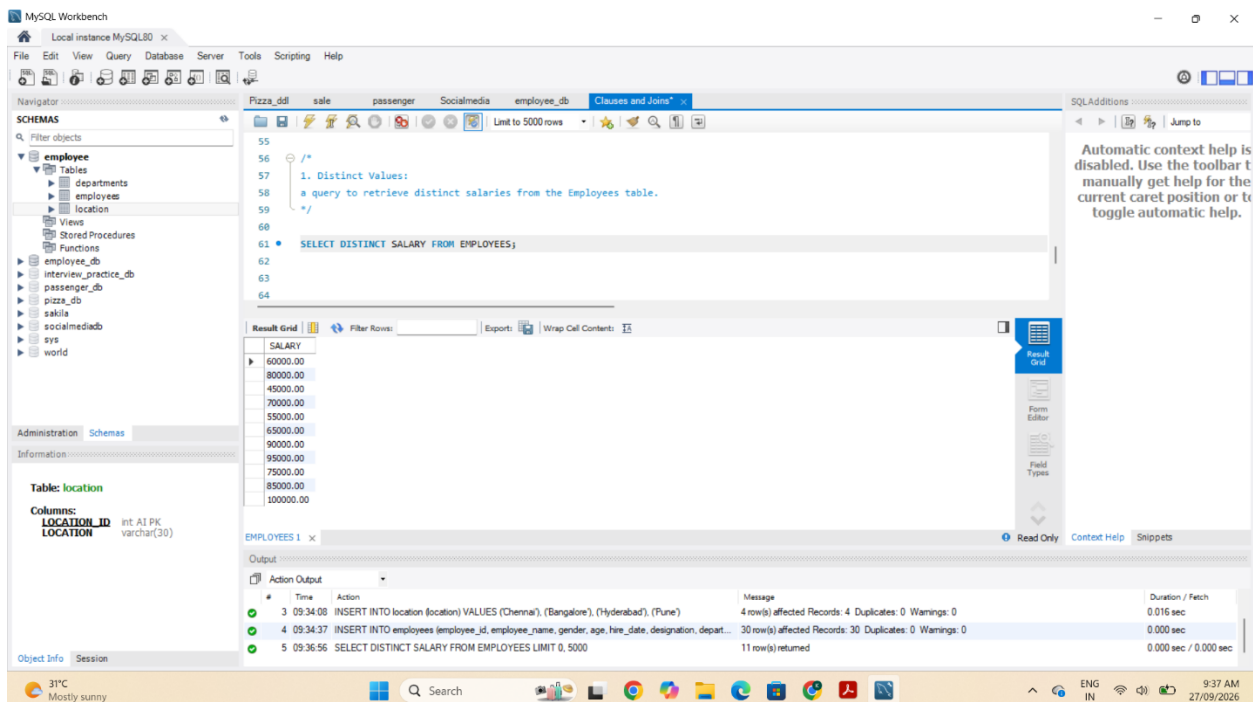
#	Time	Action	Message	Duration / Fetch
1	09:33:03	USE employee	0 row(s) affected	0.000 sec
2	09:33:37	INSERT INTO departments (department_id, department_name) VALUES (1, 'Software Development'), (2, ...	13 row(s) affected Records: 13 Duplicates: 0 Warnings: 0	0.016 sec

The Navigator panel now shows "No object selected". The Context Help panel remains the same.



1. Distinct Values:

- a query to retrieve distinct salaries from the Employees table.



2. Alias (AS):

- Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
/*
2. Alias (AS):
Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.
*/
SELECT
  AGE AS EMPLOYEE_AGE,
  SALARY AS EMPLOYEE_SALARY
FROM EMPLOYEES;
```

The result grid displays the following data:

EMPLOYEE_AGE	EMPLOYEE_SALARY
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	50000.00
32	50000.00
27	70000.00

The bottom panel shows the Action Output with the following messages:

#	Time	Action	Message	Duration / Fetch
4	09:34:37	INSERT INTO employees (employee_id, employee_name, gender, age, hire_date, designation, depart...	30 row(s) affected Records: 30 Duplicates: 0 Warnings: 0	0.000 sec
5	09:36:56	SELECT DISTINCT SALARY FROM EMPLOYEES LIMIT 0, 5000	11 row(s) returned	0.000 sec / 0.000 sec
6	09:39:48	SELECT AGE AS EMPLOYEE_AGE, SALARY AS EMPLOYEE_SALARY FROM EMPLOYEES LI...	30 row(s) returned	0.000 sec / 0.000 sec

3. Where Clause & Operators:

- Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.
- Find the employee whose designation is missing and fill it with "Data Scientist".

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee

Tables

departments

employees

Columns

EMPLOYEE_ID

EMPLOYEE_NAME

GENDER

AGE

HIRE_DATE

DESIGNATION

DEPARTMENT_ID

LOCATION_ID

SALARY

Indexes

Foreign Keys

Triggers

location

Views

Stored Procedures

Functions

Administration Schemas

Information

Table: location

Columns:

LOCATION_ID int(11) PK

LOCATION varchar(30)

Object Info Session

31°C Mostly sunny

Search

ENG IN

9:43 AM 27/09/2026

Limit to 5000 rows

SQL Editor

```

70  SALARY AS EMPLOYEE_SALARY
71  FROM EMPLOYEES;
72
73  /*
74  3. Where Clause & Operators:
75  # Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.
76  # Find the employee whose designation is missing and fill it with "Data Scientist".
77  */
78
79  SELECT * FROM employees
80  WHERE SALARY > 50000
81  AND HIRE_DATE < '2016-01-01';
82
83

```

Result Grid

EMPLOYEE_ID	EMPLOYEE_NAME	GENDER	AGE	HIRE_DATE	DESIGNATION	DEPARTMENT_ID	LOCATION_ID	SALARY
5001	Vhaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyash Singh	M	31	2015-03-30	Network Engineer	12	1	80000.00
5004	Kiana Malhotra	F	29	2015-07-05		8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00

employees 3 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	09:36:56	SELECT DISTINCT SALARY FROM EMPLOYEES LIMIT 0, 5000	11 row(s) returned	0.000 sec / 0.000 sec
6	09:39:48	SELECT AGE AS EMPLOYEE_AGE, SALARY AS EMPLOYEE_SALARY FROM EMPLOYEES U...	30 row(s) returned	0.000 sec / 0.000 sec
7	09:43:35	SELECT * FROM employees WHERE SALARY > 50000 AND HIRE_DATE < '2016-01-01' LIMIT 0, 5...	5 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee

Tables

departments

employees

Columns

EMPLOYEE_ID

EMPLOYEE_NAME

GENDER

AGE

HIRE_DATE

DESIGNATION

DEPARTMENT_ID

LOCATION_ID

SALARY

Indexes

Foreign Keys

Triggers

location

Views

Stored Procedures

Functions

Administration Schemas

Information

Table: location

Columns:

LOCATION_ID int(11) PK

LOCATION varchar(30)

Object Info Session

31°C Mostly sunny

Search

ENG IN

9:45 AM 27/09/2026

Limit to 5000 rows

SQL Editor

```

72
73  /*
74  3. Where Clause & Operators:
75  # Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.
76  # Find the employee whose designation is missing and fill it with "Data Scientist".
77  */
78
79  SELECT * FROM employees
80  WHERE SALARY > 50000
81  AND HIRE_DATE < '2016-01-01';
82
83  SELECT * FROM EMPLOYEES
84  WHERE DESIGNATION IS NULL;
85

```

Result Grid

EMPLOYEE_ID	EMPLOYEE_NAME	GENDER	AGE	HIRE_DATE	DESIGNATION	DEPARTMENT_ID	LOCATION_ID	SALARY
5004	Kiana Malhotra	F	29	2015-07-05		8	3	70000.00

EMPLOYEES 6 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	09:45:23	SELECT * FROM EMPLOYEES WHERE DESIGNATION IS NULL LIMIT 0, 5000	1 row(s) returned	0.016 sec / 0.000 sec
9	09:45:33	SELECT * FROM employees WHERE SALARY > 50000 AND HIRE_DATE < '2016-01-01' LIMIT 0, 5...	5 row(s) returned	0.000 sec / 0.000 sec
10	09:45:39	SELECT * FROM EMPLOYEES WHERE DESIGNATION IS NULL LIMIT 0, 5000	1 row(s) returned	0.016 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee

Tables

departments

employees

Columns

EMPLOYEE_ID

EMPLOYEE_NAME

GENDER

AGE

HIRE_DATE

DESIGNATION

DEPARTMENT_ID

LOCATION_ID

SALARY

Indexes

Foreign Keys

Triggers

location

Views

Stored Procedures

Functions

Administration Schemas

Information

Table: location

Columns:

LOCATION_ID int AI PK

LOCATION varchar(30)

Object Info Session

31°C Mostly sunny

Search

ENG IN

9:48 AM 27/09/2026

SQL Editor

Limit to 5000 rows

77

78

79 SELECT * FROM employees

80 WHERE SALARY > 50000

81 AND HIRE_DATE < '2016-01-01';

82

83 SELECT * FROM EMPLOYEES

84 WHERE DESIGNATION IS NULL;

85

86 UPDATE EMPLOYEES

87 SET DESIGNATION = 'Data Scientist'

88 WHERE EMPLOYEE_ID = 5004;

89 SELECT * FROM EMPLOYEES WHERE EMPLOYEE_ID = 5004;

90

Result Grid

EMPLOYEE_ID	EMPLOYEE_NAME	GENDER	AGE	HIRE_DATE	DESIGNATION	DEPARTMENT_ID	LOCATION_ID	SALARY
5004	Kiera Mahotra	F	29	2015-07-05	Data Scientist	8	3	70000.00

EMPLOYEES 7

Output

#	Time	Action	Message	Duration / Fetch
10	09:45:39	SELECT * FROM EMPLOYEES WHERE DESIGNATION IS NULL LIMIT 0, 5000	1 row(s) returned	0.016 sec / 0.000 sec
11	09:48:21	UPDATE EMPLOYEES SET DESIGNATION = 'Data Scientist' WHERE EMPLOYEE_ID = 5004	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
12	09:48:21	SELECT * FROM EMPLOYEES WHERE EMPLOYEE_ID = 5004 LIMIT 0, 5000	1 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

1. ORDER BY:

- Find employees sorted by department ID in ascending order and salary in descending order.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee

Tables

departments

employees

Columns

EMPLOYEE_ID

EMPLOYEE_NAME

GENDER

AGE

HIRE_DATE

DESIGNATION

DEPARTMENT_ID

LOCATION_ID

SALARY

Indexes

Foreign Keys

Triggers

location

Views

Stored Procedures

Functions

Administration Schemas

Information

Table: location

Columns:

LOCATION_ID int AI PK

LOCATION varchar(30)

Object Info Session

31°C Mostly sunny

Search

ENG IN

10:38 AM 27/09/2026

SQL Editor

Limit to 5000 rows

89 SELECT * FROM EMPLOYEES WHERE EMPLOYEE_ID = 5004;

90

91 -- Sorting and Grouping Data:

92 /*

93 1. ORDER BY:

94 Find employees sorted by department ID in ascending order and salary in descending order.

95 */

96

97 SELECT * FROM EMPLOYEES

98 ORDER BY DEPARTMENT_ID ASC , SALARY DESC ;

99

100

Result Grid

EMPLOYEE_ID	EMPLOYEE_NAME	GENDER	AGE	HIRE_DATE	DESIGNATION	DEPARTMENT_ID	LOCATION_ID	SALARY
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00
5011	Saamvi Patel	F	28	2016-09-20	Marketing Analyst	2	1	60000.00
5025	Aarvi Tyengar	F	31	2019-01-20	Data Scientist	3	4	100000.00
5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	3	2	75000.00
5001	Vibhaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	4	4	100000.00
5022	Aadhyia Tyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00

EMPLOYEES 8

Output

#	Time	Action	Message	Duration / Fetch
11	09:48:21	UPDATE EMPLOYEES SET DESIGNATION = 'Data Scientist' WHERE EMPLOYEE_ID = 5004	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
12	09:48:21	SELECT * FROM EMPLOYEES WHERE EMPLOYEE_ID = 5004 LIMIT 0, 5000	1 row(s) returned	0.000 sec / 0.000 sec
13	10:37:39	SELECT * FROM EMPLOYEES ORDER BY DEPARTMENT_ID ASC , SALARY DESC LIMIT 0, 5000	30 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

2. LIMIT:

- Display the first 5 employees hired in the year 2018.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
/*  
2. LIMIT:  
# Display the first 5 employees hired in the year 2018.  
*/  
  
SELECT *  
FROM EMPLOYEES  
WHERE HIRE_DATE >= '2018-01-01'  
AND HIRE_DATE < '2019-01-01'  
ORDER BY HIRE_DATE ASC  
LIMIT 5;
```

The Result Grid displays the following data:

EMPLOYEE_ID	EMPLOYEE_NAME	GENDER	AGE	HIRE_DATE	DESIGNATION	DEPARTMENT_ID	LOCATION_ID	SALARY
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	7	3	85000.00
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00

The Output panel shows the execution of the query:

Time	Action	Message	Duration / Fetch
31 16:59:59	SELECT * FROM EMPLOYEES LIMIT 0, 5000	30 row(s) returned	0.000 sec / 0.000 sec
32 17:00:28	SELECT E.EMPLOYEE_NAME, L.LOCATION_ID FROM EMPLOYEES E RIGHT JOIN LOCATION...	30 row(s) returned	0.016 sec / 0.000 sec
33 17:05:41	SELECT * FROM EMPLOYEES WHERE HIRE_DATE >= '2018-01-01' AND HIRE_DATE < '2019-01-01'...	5 row(s) returned	0.000 sec / 0.000 sec

3. Aggregate Functions:

- Calculate the sum of all salaries in the Finance department.
- Find the minimum age among all employees.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

employee

Tables

departments

Columns

DEPARTMENT_ID

DEPARTMENT_NAME

Indexes

Foreign Keys

Triggers

employees

Columns

EMPLOYEE_ID

EMPLOYEE_NAME

GENDER

AGE

HIRE_DATE

DESIGNATION

DEPARTMENT_ID

LOCATION_ID

SALARY

Indexes

Administration Schemas

Information

Column: DEPARTMENT_ID

Definition: DEPARTMENT_ID int PK

Object Info Session

31°C Mostly sunny

Search

Limit to 5000 rows

107 LIMIT 5;

108

109

110 /*

111 3. Aggregate Functions:

112 Calculate the sum of all salaries in the Finance department.

113 Find the minimum age among all employees.

114 */

115 SELECT D.DEPARTMENT_NAME, SUM(SALARY) AS TOTAL_SALARY FROM EMPLOYEES E

116 INNER JOIN DEPARTMENTS D ON E.DEPARTMENT_ID = D.DEPARTMENT_ID

117 WHERE DEPARTMENT_NAME = 'Finance';

118

Result Grid

DEPARTMENT_NAME	TOTAL_SALARY
Finance	170000.00

Result 10

Read Only

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	10:37:39	SELECT * FROM EMPLOYEES ORDER BY DEPARTMENT_ID ASC, SALARY DESC LIMIT 0, 5000	30 row(s) returned	0.000 sec / 0.000 sec
14	10:45:05	SELECT * FROM EMPLOYEES WHERE YEAR(HIRE_DATE) = 2018 LIMIT 5	5 row(s) returned	0.016 sec / 0.000 sec
15	11:32:25	SELECT D.DEPARTMENT_NAME, SUM(SALARY) AS TOTAL_SALARY FROM EMPLOYEES E INN...	1 row(s) returned	0.000 sec / 0.000 sec

ENG IN

11:32 AM

27/09/2026

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

company_db

Tables

Views

Stored Procedures

Functions

ANNUAL_SALARY

SALARY_WITH_BONUS

employee

Tables

Views

Stored Procedures

Functions

employee_db

interview_practice_db

passenger_db

pizza_db

sakila

socialmediadb

sys

world

Administration Schemas

Information

No object selected

Object Info Session

31°C Mostly sunny

Search

Limit to 5000 rows

116 Calculate the sum of all salaries in the Finance department.

117 Find the minimum age among all employees.

118 /*

119

120 SELECT D.DEPARTMENT_NAME, SUM(SALARY) AS TOTAL_SALARY FROM EMPLOYEES E

121 INNER JOIN DEPARTMENTS D ON E.DEPARTMENT_ID = D.DEPARTMENT_ID

122 WHERE DEPARTMENT_NAME = 'Finance';

123

124 SELECT MIN(AGE) AS MIN_AGE FROM EMPLOYEES;

Result Grid

MIN_AGE
25

Result 1

Read Only

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	10:11:42	USE employee	0 row(s) affected	0.000 sec
2	10:12:02	SELECT MIN(AGE) AS MIN_AGE FROM EMPLOYEES LIMIT 0, 5000	1 row(s) returned	0.000 sec / 0.000 sec

ENG IN

10:12 AM

29/09/2026

4. GROUP BY:

- List the maximum salary for each location.
- Calculate the average salary for each designation containing the word 'Analyst'.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
116 INNER JOIN DEPARTMENTS D ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
117 WHERE DEPARTMENT_NAME = 'Finance';
118
119 /*
120 4. GROUP BY:
121 List the maximum salary for each location.
122 Calculate the average salary for each designation containing the word 'Analyst'.
123 */
124
125 SELECT L.LOCATION, MAX(SALARY) AS MAX_SALARY FROM EMPLOYEES E
126 INNER JOIN LOCATION L ON E.LOCATION_ID = L.LOCATION_ID
127 GROUP BY L.LOCATION;
```

The Results window shows the output of the query:

LOCATION	MAX_SALARY
Bangalore	95000.00
Chennai	95000.00
Hyderabad	95000.00
Pune	100000.00

The Output window shows the execution details:

#	Time	Action	Message	Duration / Fetch
17	11:33:28	SELECT * FROM EMPLOYEES WHERE DEPARTMENT_ID = 7 LIMIT 0, 5000	2 row(s) returned	0.016 sec / 0.000 sec
18	11:33:57	SELECT D.DEPARTMENT_NAME, SUM(SALARY) AS TOTAL_SALARY FROM EMPLOYEES E INNER JOIN DEPARTMENTS D ON E.DEPARTMENT_ID = D.DEPARTMENT_ID WHERE DEPARTMENT_NAME = 'Finance';	1 row(s) returned	0.000 sec / 0.000 sec
19	12:53:08	SELECT L.LOCATION, MAX(SALARY) AS MAX_SALARY FROM EMPLOYEES E INNER JOIN LOCATION L ON E.LOCATION_ID = L.LOCATION_ID GROUP BY L.LOCATION;	4 row(s) returned	0.015 sec / 0.000 sec

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
121 List the maximum salary for each location.
122 Calculate the average salary for each designation containing the word 'Analyst'.
123 */
124
125 SELECT L.LOCATION, MAX(SALARY) AS MAX_SALARY FROM EMPLOYEES E
126 INNER JOIN LOCATION L ON E.LOCATION_ID = L.LOCATION_ID
127 GROUP BY L.LOCATION;
128
129 SELECT DESIGNATION, AVG(SALARY) AS AVG_SALARY FROM EMPLOYEES
130 GROUP BY DESIGNATION
131 HAVING DESIGNATION LIKE 'Analyst%';
132
```

The Results window shows the output of the query:

DESIGNATION	AVG_SALARY
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

The Output window shows the execution details:

#	Time	Action	Message	Duration / Fetch
19	12:53:08	SELECT L.LOCATION, MAX(SALARY) AS MAX_SALARY FROM EMPLOYEES E INNER JOIN LOCATION L ON E.LOCATION_ID = L.LOCATION_ID GROUP BY L.LOCATION;	4 row(s) returned	0.015 sec / 0.000 sec
20	12:56:47	SELECT DESIGNATION, AVG(SALARY) FROM EMPLOYEES GROUP BY DESIGNATION HAVING DESIGNATION LIKE 'Analyst%';	7 row(s) returned	0.000 sec / 0.000 sec
21	12:58:19	SELECT DESIGNATION, AVG(SALARY) AS AVG_SALARY FROM EMPLOYEES GROUP BY DESIGNATION HAVING DESIGNATION LIKE 'Analyst%';	7 row(s) returned	0.000 sec / 0.000 sec

5. HAVING:

- Find departments with less than 3 employees.
- Find locations with female employees whose average age is below 30.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the database schema, including tables like departments and employees. The main editor window contains a SQL query:

```
130 WHERE DESIGNATION LIKE '%Analyst%'
131 GROUP BY DESIGNATION;
132
133 /*
134 5. HAVING:
135 Find departments with less than 3 employees.
136 Find locations with female employees whose average age is below 30.
137 */
138
139 SELECT DEPARTMENT_ID, COUNT(*) AS EMP_COUNT FROM EMPLOYEES
140 GROUP BY DEPARTMENT_ID
141 HAVING EMP_COUNT < 3;
```

The query results are displayed in a table with two columns: DEPARTMENT_ID and EMP_COUNT.

DEPARTMENT_ID	EMP_COUNT
1	1
5	2
6	2
7	2
9	1
11	2
13	1

The bottom status bar shows the system time as 2:39 PM on 27/09/2026.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the database schema, including tables like departments and employees. The main editor window contains a SQL query:

```
135 Find departments with less than 3 employees.
136 Find locations with female employees whose average age is below 30.
137 */
138
139 SELECT DEPARTMENT_ID, COUNT(*) AS EMP_COUNT FROM EMPLOYEES
140 GROUP BY DEPARTMENT_ID
141 HAVING EMP_COUNT < 3;
142
143 SELECT LOCATION_ID, AVG(AGE) AS AVG_AGE FROM EMPLOYEES
144 WHERE GENDER = 'F'
145 GROUP BY LOCATION_ID
146 HAVING AVG_AGE < 30;
```

The query results are displayed in a table with two columns: LOCATION_ID and AVG_AGE.

LOCATION_ID	AVG_AGE
1	26.7500
2	27.3333
4	29.2000

The bottom status bar shows the system time as 2:41 PM on 27/09/2026.

Joins:

1. Inner Join:

- List employee names, their designations, and department names where employees are assigned to a department.

The screenshot shows the MySQL Workbench interface. The 'Schemas' pane on the left displays the 'employee' database structure, including tables 'departments' and 'employees'. The 'SQL Editor' pane in the center contains the following SQL query:

```
-- Joins:
1. Inner Join:
-- List employee names, their designations, and department names where employees are assigned to a department.
SELECT
  E.EMPLOYEE_NAME,
  E.DESIGNATION,
  D.DEPARTMENT_NAME
FROM EMPLOYEES E
INNER JOIN DEPARTMENTS D ON E.DEPARTMENT_ID = D.DEPARTMENT_ID;
```

The 'Result Grid' pane at the bottom displays the query results in a table with three columns: EMPLOYEE_NAME, DESIGNATION, and DEPARTMENT_NAME. The results are as follows:

EMPLOYEE_NAME	DESIGNATION	DEPARTMENT_NAME
Anil Chaudhary	Business Development Executive	Business Development
Kaani Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Ananya Kapoor	Event Coordinator	Content Creation
Aaradhya Tyer	Customer Support Executive	Customer Support
Arnav Rao	Customer Success Manager	Customer Support
Vishal Mohan	Supply Chain Analyst	Customer Support
Vishal Singh	Data Analyst	Data Science
Arvi Iyengar	Data Scientist	Data Science

The 'Output' pane at the bottom shows the execution log with three entries, all successful:

#	Time	Action	Message	Duration / Fetch
23	14:38:44	SELECT DEPARTMENT_ID, COUNT(*) AS EMP_COUNT FROM EMPLOYEES GROUP BY DEPAR...	7 row(s) returned	0.000 sec / 0.000 sec
24	14:41:40	SELECT LOCATION_ID, AVG(AGE) AS AVG_AGE FROM EMPLOYEES WHERE GENDER = 'F' GR...	3 row(s) returned	0.000 sec / 0.000 sec
25	14:46:09	SELECT E.EMPLOYEE_NAME, E.DESIGNATION, D.DEPARTMENT_NAME FROM EMPLOYEE...	30 row(s) returned	0.000 sec / 0.000 sec

2. Left Join:

- List all departments along with the total number of employees in each department, including departments with no employees.

MySQL Workbench interface showing a query in the 'Clauses and Joins' tab. The query is a Left Join between EMPLOYEES (E) and DEPARTMENTS (D) on the condition E.DEPARTMENT_ID = D.DEPARTMENT_ID. The result grid displays the following data:

DEPARTMENT_NAME	EMP_COUNT
Business Development	2
Content Creation	2
Customer Support	3
Data Science	3
Design	3
Finance	2
Human Resources	3
IT	4
Marketing	3

The output pane shows the execution of the query, with a message indicating 13 rows returned.

3. Right Join:

- Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

MySQL Workbench interface showing a query in the 'Clauses and Joins' tab. The query is a Right Join between EMPLOYEES (E) and LOCATION (L) on the condition E.LOCATION_ID = L.LOCATION_ID. The result grid displays the following data:

EMPLOYEE_NAME	LOCATION
Aaradhya Iyer	Bangalore
Dhruv Shetty	Bangalore
Myra Verma	Bangalore
Vhaan Mohan	Bangalore
Kabr Nair	Bangalore
Anika Paul	Bangalore
Vivaan Singh	Bangalore
Arijun Kumar	Bangalore
Reyansh Singh	Chennai

The output pane shows the execution of the query, with a message indicating 30 rows returned.